Ridge and LASSO

1. Shrinkage methods

Fit a model containing all p predictors using a technique that constrains or regularizes the coefficient estimates, or equivalently, that shrinks the coefficient estimates towards zero.

It turns out that shrinking the coefficient estimates can significantly improve the modeling fitting by **reducing the variance** (Ridge regression, LASSO, etc) and **performing variable selection** (LASSO, Group LASSO, SCAD, MCP, etc).

In general, shrinkage methods obtain coefficient estimates by minimizing an objective function of the form:

$$\mathcal{L}(\beta) + P(\beta;\ \lambda) \iff \text{Loss function} + \text{Penalty on } \beta$$

- **Loss function**: measures the goodness-of-fit of the model with the given β value.
- **Penalty on** β: put constraints on β to achieve certain effects (e.g. shrinkage, variable selection, etc)

By using different loss functions and penalty functions, we can perform different shrinkage estimations on different (parametric) models.

2. Ridge Regression

2.1 Motivation

In practice, it is not unusual to encounter a real problem in which:

1. there are too many predictors
    1. E.g., in the high-dimension setting where n < p.
    2. The OLS is not applicable in this case.
2. the design matrix X is ill-conditioned
    1. E.g., if the predictors are high-correlated (i.e. severe multi-colinearity)
    2. $X^T X$ is singular or nearly singular
    3. The variance of the OLS estimator would be very large, i.e., asmall change in data would lead to a big change in $\hat{\beta}$
    4. The prediction may be very inaccurate in this case

2.2 Formulation

Recall that the ordinary least squares fitting procedure estimates $\boldsymbol{\beta} = (\beta_0, \cdots, \beta_p)^T$ using the values that minimize:

$$RSS = \sum_{i=0}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2$$

In contrast, the ridge regression coefficient estimates $\hat{\boldsymbol{\beta}}_\lambda^R$ are the values that minimize:

$$\sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^{p} \beta_j^2 = RSS + \lambda \sum_{j=1}^{p} \beta_j^2$$

where $\lambda \geq 0$ is a tuning parameter, to be determined separately.

The objective function of ridge regression consists of two parts:

1. **RSS**: as with OLS, ridge regression seeks coefficient estimates that fit the data well, by making the RSS small.
2. **Shrinkage penalty** (**squared l_2 norm of $\boldsymbol{\beta}$**): **shrinking** the estimates of β_j towards zero. (Note that β_0 is not penalized).

The tuning parameter λ serves to control the relative impact of these two terms on the regression coefficient estimates.

Ridge regression will produce a different set of coefficient estimates, $\hat{\boldsymbol{\beta}}_\lambda^R$, for each value of λ.

Selecting a good value for λ is critical; cross-validation is used for this.

2.3 Standardization of data

- **Centering**: recall the ridge regression does not penalize the intercept β_0. We can centering Y and $X_1, \cdots, X_p$ so that the model does not involve an intercept.
- **Scaling**: the ridge regression coefficient estimates can change substantially when multiplying a given predictor by a constant, due to the penalty term $\lambda \sum_{j=1}^{p} \beta_j^2$. We need to scale the predictors $X_1, \cdots, X_p$ so that each column of X has variance 1.
- Typically, we standardize the data before doing ridge regression as follows:

$$\tilde{y}_i = y_i - \overline{y}$$

$$\tilde{x}_{ij} = \frac{x_{ij} - \overline{x}_j}{\sqrt{\frac{1}{n} \sum_{i=1}^{n} (x_{ij} - \overline{x}_j)^2}}$$

2.4 Derivation

Suppose that $\mathbf{y}$ and $\mathbf{X}$ are standardized as mentioned before. We aim to estimate the model $\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$ via ridge regression.

The ridge estimator is obtained by:

$$\begin{aligned}\hat{\boldsymbol{\beta}}_{\lambda}^{R} &= arg\ \min_{\boldsymbol{\beta}} ||\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}||_2^2 + \lambda||\boldsymbol{\beta}||_2^2 \\ &= arg\ \min_{\boldsymbol{\beta}} (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})^T(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}) + \lambda\boldsymbol{\beta}^T\boldsymbol{\beta}\end{aligned}$$

$$\hat{\boldsymbol{\beta}}_{\lambda}^{R} = (\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^T\mathbf{y}$$

2.5 Selecting the tuning parameter λ

As for subset selection, for ridge regression and LASSO we require a method to determine which of the models under consideration is best.

Cross-validation procedure:

1. Choose a grid of λ values, and compute the cross-validation error rate for each value of λ.
2. Select the tuning parameter value for which the cross-validation error is smallest.
3. Refit the model using all of the available observations and the selected value of the tuning parameter.

3. Least Absolute Shrinkage and Selection Operator (LASSO)

3.1 Motivation

One obvious disadvantage of Ridge regression: no variable selection. The penalty $\lambda \sum_{j=1}^{p} \beta_j^2$ will shrink all of the coefficients towards zero, but it will not set any of them exactly to zero (unless $\lambda = \infty$).

This may not be a problem for prediction accuracy, but it can create a challenge in **model interpretation** when the number of predictors p is quite large.

The **Least Absolute Shrinkage and Selection Operator (LASSO)** (Tibshirani, 1996) is a relatively recent alternative to Ridge regression that overcomes this disadvantage.

3.2 Formulation

The LASSO estimates $\hat{\boldsymbol{\beta}}_{\lambda}^{L}$ are the values that minimize:

$$\sum_{i=1}^{n}\left(y_i-\beta_0-\sum_{j=1}^{p}\beta_j x_{ij}\right)^2+\lambda\sum_{j=1}^{p}|\beta_j|=RSS+\lambda\sum_{j=1}^{p}|\beta_j|$$

where $\lambda \geq 0$ is a tuning parameter, to be determined separately.

In statistical parlance, the LASSO uses an l_1 penalty instead of an l_2 penalty. The l_1 norm of a coefficient vector $\boldsymbol{\beta}$ is $||\boldsymbol{\beta}||_1=\sum|\beta_j|$.

As with ridge regression, the LASSO shrinks the coefficient estimates towards zero.

However, in the case of the LASSO, the l_1 penalty has the effect of **forcing some of the coefficient estimates to be exactly equal to zero** when the tuning parameter λ is sufficiently large.

Hence, much like best subset selection, the LASSO performs **variable selection**.

We say that the LASSO yields **sparse** models — that is, models that involve only a subset of the variables.

We need **standardization** before performing LASSO regression.

We use **cross-validation** to select the tuning parameter λ.

3.3 Computation Algorithm

Suppose the data are standardized so that there is no intercept term. The objective function for obtaining a LASSO estimate is:

$$\mathcal{L}(\boldsymbol{\beta})=\frac{1}{2}\sum_{i=1}^{n}\left(y_i-\sum_{j=1}^{p}\beta_j x_{ij}\right)^2+\lambda\sum_{j=1}^{p}|\beta_j|$$

It can be shown that keeping $\beta_l, l\neq j$ fixed and updating β_j, the formula is given by:

$$\beta_j^{(new)}=sign\left(\frac{\sum_{i=1}^{n}r_i x_{ij}}{\sum_{i=1}^{n}x_{ij}^2}\right)\max\left(\frac{|\sum_{i=1}^{n}r_i x_{ij}|}{\sum_{i=1}^{n}x_{ij}^2}-\frac{\lambda}{\sum_{i=1}^{n}x_{ij}^2},0\right)$$

where $r_i=y_i-\sum_{l\neq j}x_{il}\beta_l$.

If we define $S(x;\lambda)=sign(x)\max(x-\lambda,0)$, then the solution can be compactly written as:

$$\beta_j^{(new)}=S\left(\frac{\sum_{i=1}^{n}r_i x_{ij}}{\sum_{i=1}^{n}x_{ij}^2};\frac{\lambda}{\sum_{i=1}^{n}x_{ij}^2}\right)$$

The function $S(x;\lambda)$ is called **soft-thresholding operator**, which is very popular in high-dimensional statistical learning algorithms.

3.4 Selecting the tuning parameter λ

This is the same as the procedure of Ridge Regression.

4. A comparison of the LASSO and ridge regression

Question: Why is it that the LASSO, unlike ridge regression, results in coefficient estimates that are exactly equal to zero?

Another formulation of ridge regression and the LASSO:
Ridge Regression:

$$\min_{\beta} \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right) \quad \text{subject to} \sum_{j=1}^{p} \beta_j^2 \leq s$$

LASSO:

$$\min_{\beta} \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right) \quad \text{subject to} \sum_{j=1}^{p} |\beta_j| \leq s$$

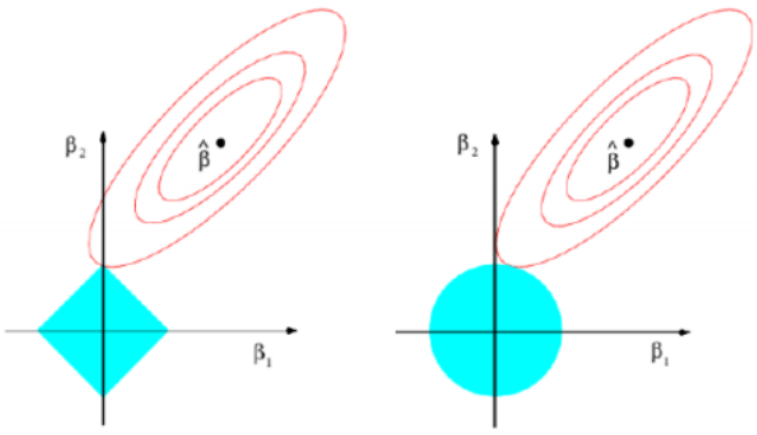

Figure: Contours of the error and constraint functions for the LASSO (left) and ridge regression (right). The solid blue areas are the constraint regions, $|\beta_1| + |\beta_2| \leq s$ and $\beta_1^2 + \beta_2^2 \leq s$, while the red ellipses are the contours of the RSS.

As the figure showed above, the contours of RSS may be tangent to the constrant regions at exactly $\beta_1 = 0$ in the case of LASSO.

Important Remarks:

- Neither ridge regression nor the LASSO will universally dominate the other.
- In general, one might expect the LASSO to perform better when the response is a function of only a relatively small number of predictors (i.e., sparsity assumption is satisfied).
- However, the number of predictors that is related to the response is never known **a priori** for real data sets.
- A technique such as cross-validation can be used in order to determine which approach is better on a particular data set.